No.	Status	Date
LNDE-FN-20241025-031-DS-4-RA-101	Issued	2024-11-07

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-101, located within the ATU-100. The objective is to assess the current wall thickness against the minimum allowable thickness to determine the remaining life and next inspection interval.

Marcus Chen, acting as the lead UT technician, performed the measurements and data analysis. All field activities were conducted in accordance with the requirements of QSCP-NDT-001 Rev 3 and the guidelines specified in ASTM E797.

Technical Specifications

The subject line, 4-RA-101, is constructed from A106 Gr.B with a nominal wall thickness of 8.56 mm and an outer diameter of 114.3 mm. The design parameters for this rich amine service are 862 kPa(g) and 82 °C.

The retirement thickness for this asset is defined as the t-min of 3.81 mm. Any measurement recorded below this threshold is classified as an NCR.

Measurement Methodology

Thickness data was collected as part of the survey detailed in LNDE-FN-20241025-031. The inspection focused on identified CMLs to monitor wall loss and calculate corrosion rates based on the larger of the long-term or short-term rates.

The data acquisition followed the digital export and CML-id formatting requirements mandated by QSCP-NDT-001 Rev 3 to ensure consistency with the site asset integrity database.

Analysis of Findings

Several CMLs on 4-RA-101 have been identified as expired. Specifically, CML 4-RA-101-06 shows a current wall thickness of 3.59 mm, which is below the t-min of 3.81 mm.

Additional critical readings were recorded at CML 2-RF-102-04 (1.94 mm), 2-RF-102-05 (1.92 mm), 2-RF-102-06 (1.92 mm), and 2-RF-102-07 (1.86 mm). All these locations are reported as expired.

Integrity Status

Based on the recorded thicknesses and the t-min of 3.81 mm, the remaining life for the aforementioned CMLs has been exhausted. These readings constitute a failure to meet the minimum thickness requirements.

The identified wall loss at these locations requires immediate attention as they have fallen below the retirement thickness specified for the line.

Thickness readings, measured 2024-11-04

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2022-09-16	5.82	5.64	0.052	0.057	36.2	2027-11-04	
6-SG-101-02	7.11	3.56	2022-09-16	5.42	5.12	0.067	0.077	20.2	2027-11-04	
6-SG-101-03	7.11	3.56	2022-09-16	5.29	5.25	0.064	0.061	26.5	2027-11-04	
6-SG-101-04	7.11	3.56	2022-09-16	5.19	4.98	0.074	0.071	19.3	2027-11-04	
6-SG-101-05	7.11	3.56	2022-09-16	4.72	4.65	0.087	0.077	12.5	2027-11-04	
6-SG-101-06	7.11	3.56	2022-09-16	5.55	5.39	0.059	0.034	31.2	2027-11-04	
6-SG-101-07	7.11	3.56	2022-09-16	6.16	6.04	0.040	0.040	>50	2027-11-04	
6-SG-101-08	7.11	3.56	2022-09-16	5.57	5.54	0.058	0.046	34.4	2027-11-04	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2022-09-16	4.77	4.62	0.088	0.085	12.1	2027-11-04	
6-SG-102-02	7.11	3.56	2022-09-16	4.64	4.59	0.088	0.101	10.2	2027-11-04	
6-SG-102-03	7.11	3.56	2022-09-16	4.69	4.45	0.090	0.093	9.6	2027-11-04	gauge photo attached
6-SG-102-04	7.11	3.56	2022-09-16	4.62	4.39	0.091	0.101	8.2	2027-11-04	gauge photo attached
6-SG-102-05	7.11	3.56	2022-09-16	5.96	5.89	0.043	0.036	>50	2027-11-04	
6-SG-102-06	7.11	3.56	2022-09-16	5.79	5.82	0.043	0.042	>50	2027-11-04	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2019-10-18	7.63	7.57	0.022	0.012	>50	2030-11-04	
8-TG-101-02	8.18	4.06	2019-10-18	7.33	7.21	0.035	0.024	>50	2030-11-04	
8-TG-101-03	8.18	4.06	2019-10-18	7.37	7.27	0.033	0.020	>50	2030-11-04	
8-TG-101-04	8.18	4.06	2019-10-18	7.38	7.29	0.028	0.018	>50	2030-11-04	
8-TG-101-05	8.18	4.06	2019-10-18	7.57	7.54	0.024	0.006	>50	2030-11-04	
8-TG-101-06	8.18	4.06	2019-10-18	7.75	7.61	0.028	0.028	>50	2030-11-04	
8-TG-101-07	8.18	4.06	2019-10-18	7.17	6.98	0.047	0.038	>50	2030-11-04	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2019-10-18	4.70	4.46	0.053	0.048	24.0	2030-11-04	
4-LA-101-02	6.02	3.18	2019-10-18	5.26	5.23	0.027	0.006	>50	2030-11-04	
4-LA-101-03	6.02	3.18	2019-10-18	5.38	5.19	0.027	0.038	>50	2030-11-04	
4-LA-101-04	6.02	3.18	2019-10-18	5.29	5.13	0.033	0.032	>50	2030-11-04	
4-LA-101-05	6.02	3.18	2019-10-18	5.38	5.25	0.027	0.026	>50	2030-11-04	
4-LA-101-06	6.02	3.18	2019-10-18	5.22	4.91	0.040	0.061	28.2	2030-11-04	
4-LA-101-07	6.02	3.18	2019-10-18	4.86	4.49	0.052	0.073	17.9	2030-11-04	
4-LA-101-08	6.02	3.18	2019-10-18	5.04	4.70	0.067	0.067	22.6	2030-11-04	
4-LA-101-09	6.02	3.18	2019-10-18	4.86	4.57	0.049	0.057	24.2	2030-11-04	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2019-10-18	5.09	5.03	0.035	0.012	>50	2030-11-04	
4-LA-102-02	6.02	3.18	2019-10-18	5.15	5.08	0.035	0.014	>50	2030-11-04	
4-LA-102-03	6.02	3.18	2019-10-18	5.15	4.93	0.040	0.044	40.2	2030-11-04	
4-LA-102-04	6.02	3.18	2019-10-18	4.65	4.40	0.055	0.050	22.0	2030-11-04	
4-LA-102-05	6.02	3.18	2019-10-18	5.23	5.06	0.033	0.034	>50	2030-11-04	
4-LA-102-06	6.02	3.18	2019-10-18	5.52	5.42	0.020	0.020	>50	2030-11-04	
4-LA-102-07	6.02	3.18	2019-10-18	5.32	5.13	0.030	0.038	>50	2030-11-04	
4-LA-102-08	6.02	3.18	2019-10-18	5.55	5.38	0.037	0.034	>50	2030-11-04	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2019-10-18	6.40	6.27	0.026	0.026	>50	2030-11-04	
6-LA-103-02	7.11	3.56	2019-10-18	6.49	6.28	0.032	0.042	>50	2030-11-04	
6-LA-103-03	7.11	3.56	2019-10-18	6.23	5.95	0.039	0.055	43.1	2030-11-04	
6-LA-103-04	7.11	3.56	2019-10-18	5.85	5.64	0.053	0.042	39.0	2030-11-04	
6-LA-103-05	7.11	3.56	2019-10-18	5.81	5.43	0.061	0.075	24.8	2030-11-04	
6-LA-103-06	7.11	3.56	2019-10-18	5.46	5.10	0.072	0.071	21.4	2030-11-04	
6-LA-103-07	7.11	3.56	2019-10-18	5.93	5.73	0.051	0.040	42.9	2030-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-08	7.11	3.56	2019-10-18	6.19	—	—	—	—	—	inaccessible – no reading
6-LA-103-09	7.11	3.56	2019-10-18	6.16	5.92	0.048	0.048	49.6	2030-11-04	
6-LA-103-10	7.11	3.56	2019-10-18	6.00	5.74	0.064	0.051	34.0	2030-11-04	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2022-09-16	6.83	6.50	0.070	0.156	17.2	2027-11-04	
4-RA-101-02	8.56	3.81	2022-09-16	6.79	6.46	0.073	0.184	14.4	2027-11-04	
4-RA-101-03	8.56	3.81	2022-09-16	5.98	5.52	0.108	0.222	7.7	2027-11-04	gauge photo attached
4-RA-101-04	8.56	3.81	2022-09-16	7.21	6.90	0.060	0.127	24.4	2027-11-04	
4-RA-101-05	8.56	3.81	2022-09-16	6.73	6.37	0.078	0.160	16.0	2027-11-04	
4-RA-101-06	8.56	3.81	2019-10-18	6.94	3.59	0.427	0.664	expired	immediate	< t-min; gauge photo attached
4-RA-101-07	8.56	3.81	2022-09-16	7.50	7.21	0.048	0.093	36.5	2027-11-04	
4-RA-101-08	8.56	3.81	2022-09-16	5.55	5.06	0.126	0.275	4.5	2027-02-10	gauge photo attached
4-RA-101-09	8.56	3.81	2022-09-16	7.37	7.17	0.049	0.101	33.3	2027-11-04	
4-RA-101-10	8.56	3.81	2022-09-16	7.19	6.91	0.061	0.127	24.5	2027-11-04	
4-RA-101-11	8.56	3.81	2022-09-16	6.07	5.53	0.110	0.224	7.7	2027-11-04	gauge photo attached
4-RA-101-12	8.56	3.81	2022-09-16	6.41	6.10	0.184	0.184	12.4	2027-11-04	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2022-09-16	7.42	7.33	0.042	0.050	>50	2027-11-04	
4-RA-102-02	8.56	3.81	2022-09-16	6.43	5.99	0.090	0.192	11.3	2027-11-04	
4-RA-102-03	8.56	3.81	2022-09-16	6.56	6.24	0.081	0.160	15.1	2027-11-04	
4-RA-102-04	8.56	3.81	2022-09-16	6.64	6.38	0.075	0.131	19.7	2027-11-04	
4-RA-102-05	8.56	3.81	2022-09-16	6.56	6.23	0.081	0.164	14.7	2027-11-04	
4-RA-102-06	8.56	3.81	2022-09-16	6.65	6.35	0.075	0.158	16.0	2027-11-04	
4-RA-102-07	8.56	3.81	2022-09-16	6.85	6.53	0.072	0.153	17.8	2027-11-04	
4-RA-102-08	8.56	3.81	2022-09-16	5.86	5.26	0.113	0.226	6.4	2027-11-04	gauge photo attached
4-RA-102-09	8.56	3.81	2022-09-16	6.42	6.04	0.088	0.147	15.2	2027-11-04	
4-RA-102-10	8.56	3.81	2022-09-16	6.86	6.56	0.158	0.158	17.4	2027-11-04	
4-RA-102-11	8.56	3.81	2022-09-16	7.33	7.19	0.073	0.105	32.2	2027-11-04	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2022-09-16	9.55	9.28	0.060	0.109	43.2	2027-11-04	
6-RA-103-02	10.97	4.57	2022-09-16	7.97	7.33	0.131	0.254	10.9	2027-11-04	
6-RA-103-03	10.97	4.57	2022-09-16	8.95	8.44	0.089	0.208	18.6	2027-11-04	
6-RA-103-04	10.97	4.57	2022-09-16	7.30	6.60	0.153	0.321	6.3	2027-11-04	gauge photo attached
6-RA-103-05	10.97	4.57	2022-09-16	7.43	6.91	0.141	0.281	8.3	2027-11-04	gauge photo attached
6-RA-103-06	10.97	4.57	2022-09-16	9.45	9.16	0.065	0.125	36.8	2027-11-04	
6-RA-103-07	10.97	4.57	2022-09-16	9.15	8.89	0.073	0.139	31.2	2027-11-04	
6-RA-103-08	10.97	4.57	2022-09-16	9.29	9.07	0.066	0.131	34.4	2027-11-04	
6-RA-103-09	10.97	4.57	2022-09-16	9.43	9.10	0.067	0.135	33.6	2027-11-04	
6-RA-103-10	10.97	4.57	2022-09-16	7.68	7.00	0.140	0.271	9.0	2027-11-04	gauge photo attached
6-RA-103-11	10.97	4.57	2019-10-18	8.91	5.92	0.413	0.592	2.3	2025-12-25	gauge photo attached
6-RA-103-12	10.97	4.57	2022-09-16	9.11	8.74	0.154	0.154	27.0	2027-11-04	
6-RA-103-13	10.97	4.57	2022-09-16	9.13	8.92	0.112	0.125	34.9	2027-11-04	
6-RA-103-14	10.97	4.57	2022-09-16	7.34	6.52	0.345	0.345	5.7	2027-09-02	gauge photo attached

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2022-09-16	5.92	5.74	0.065	0.061	33.8	2027-11-04	
3-RV-101-02	7.62	3.56	2022-09-16	5.81	5.70	0.065	0.053	32.8	2027-11-04	
3-RV-101-03	7.62	3.56	2022-09-16	4.98	4.72	0.098	0.103	11.3	2027-11-04	
3-RV-101-04	7.62	3.56	2022-09-16	5.39	5.18	0.084	0.079	19.3	2027-11-04	
3-RV-101-05	7.62	3.56	2022-09-16	5.91	5.84	0.064	0.053	35.7	2027-11-04	
3-RV-101-06	7.62	3.56	2022-09-16	5.39	5.22	0.083	0.065	20.1	2027-11-04	
3-RV-101-07	7.62	3.56	2022-09-16	4.80	4.63	0.111	0.087	9.6	2027-11-04	gauge photo attached
3-RV-101-08	7.62	3.56	2022-09-16	5.81	5.64	0.076	0.044	27.5	2027-11-04	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2022-09-16	3.91	3.76	0.062	0.079	12.2	2027-11-04	
2-RV-102-02	5.54	2.79	2022-09-16	3.64	3.49	0.073	0.073	9.6	2027-11-04	gauge photo attached
2-RV-102-03	5.54	2.79	2022-09-16	4.34	4.23	0.044	0.028	33.0	2027-11-04	
2-RV-102-04	5.54	2.79	2022-09-16	4.47	4.39	0.042	0.040	38.5	2027-11-04	
2-RV-102-05	5.54	2.79	2022-09-16	4.59	4.57	0.033	0.042	42.8	2027-11-04	
2-RV-102-06	5.54	2.79	2022-09-16	4.05	3.94	0.062	0.065	17.6	2027-11-04	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2019-10-18	5.40	5.15	0.012	0.050	>50	2030-11-04	
3-RF-101-02	5.49	2.41	2019-10-18	5.38	5.24	0.007	0.028	>50	2030-11-04	
3-RF-101-03	5.49	2.41	2019-10-18	5.35	5.35	0.003	0.000	>50	2030-11-04	
3-RF-101-04	5.49	2.41	2019-10-18	5.29	5.21	0.009	0.016	>50	2030-11-04	
3-RF-101-05	5.49	2.41	2019-10-18	5.17	5.14	0.011	0.006	>50	2030-11-04	
3-RF-101-06	5.49	2.41	2019-10-18	5.12	5.13	0.014	-0.002	>50	2030-11-04	
3-RF-101-07	5.49	2.41	2019-10-18	5.38	5.25	0.006	0.026	>50	2030-11-04	
3-RF-101-08	5.49	2.41	2019-10-18	5.23	5.12	0.022	0.022	>50	2030-11-04	
3-RF-101-09	5.49	2.41	2019-10-18	5.13	5.10	0.002	0.006	>50	2030-11-04	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2019-10-18	3.66	3.55	0.013	0.022	>50	2030-11-04	
2-RF-102-02	3.91	2.16	2019-10-18	3.65	3.53	0.012	0.024	>50	2030-11-04	
2-RF-102-03	3.91	2.16	2019-10-18	3.68	3.65	0.010	0.006	>50	2030-11-04	
2-RF-102-04	3.91	2.16	2019-10-18	2.83	1.94	0.067	0.176	expired	immediate	< t-min; gauge photo attached
2-RF-102-05	3.91	2.16	2019-10-18	2.88	1.92	0.159	0.190	expired	immediate	< t-min; gauge photo attached
2-RF-102-06	3.91	2.16	2019-10-18	2.74	1.92	0.160	0.162	expired	immediate	< t-min; gauge photo attached
2-RF-102-07	3.91	2.16	2019-10-18	2.82	1.86	0.073	0.190	expired	immediate	< t-min; gauge photo attached

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2019-10-18	5.16	5.02	0.037	0.028	>50	2030-11-04	
4-LA-104-02	6.02	3.18	2019-10-18	4.79	4.55	0.052	0.048	26.5	2030-11-04	
4-LA-104-03	6.02	3.18	2019-10-18	5.23	5.20	0.027	0.006	>50	2030-11-04	
4-LA-104-04	6.02	3.18	2019-10-18	5.49	5.34	0.025	0.030	>50	2030-11-04	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-05	6.02	3.18	2019-10-18	4.71	4.37	0.058	0.067	17.7	2030-11-04	
4-LA-104-06	6.02	3.18	2019-10-18	5.38	5.17	0.032	0.042	47.8	2030-11-04	
4-LA-104-07	6.02	3.18	2019-10-18	5.15	4.92	0.038	0.046	38.2	2030-11-04	
4-LA-104-08	6.02	3.18	2019-10-18	4.88	4.63	0.046	0.050	29.3	2030-11-04	
4-LA-104-09	6.02	3.18	2019-10-18	4.61	4.37	0.048	0.048	25.0	2030-11-04	
4-LA-104-10	6.02	3.18	2019-10-18	4.63	4.32	0.060	0.061	18.6	2030-11-04	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2022-09-16	6.61	6.33	0.076	0.162	15.5	2027-11-04	
4-RA-104-02	8.56	3.81	2022-09-16	7.04	6.84	0.060	0.093	32.5	2027-11-04	
4-RA-104-03	8.56	3.81	2022-09-16	7.70	7.67	0.065	0.053	>50	2027-11-04	
4-RA-104-04	8.56	3.81	2022-09-16	7.31	6.98	0.109	0.131	24.2	2027-11-04	
4-RA-104-05	8.56	3.81	2022-09-16	7.12	6.74	0.119	0.164	17.8	2027-11-04	
4-RA-104-06	8.56	3.81	2022-09-16	6.70	6.41	0.076	0.143	18.2	2027-11-04	
4-RA-104-07	8.56	3.81	2022-09-16	6.04	5.61	0.104	0.210	8.6	2027-11-04	gauge photo attached
4-RA-104-08	8.56	3.81	2022-09-16	7.09	6.90	0.093	0.111	27.9	2027-11-04	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2019-10-18	3.59	3.42	0.018	0.034	33.6	2030-11-04	
2-ST-101-02	3.91	2.29	2019-10-18	3.34	3.19	0.025	0.030	30.3	2030-11-04	
2-ST-101-03	3.91	2.29	2019-10-18	3.68	—	—	—	—	—	inaccessible – no reading
2-ST-101-04	3.91	2.29	2019-10-18	3.52	3.50	0.011	0.004	>50	2030-11-04	
2-ST-101-05	3.91	2.29	2019-10-18	3.27	3.22	0.024	0.010	38.6	2030-11-04	
2-ST-101-06	3.91	2.29	2019-10-18	3.66	3.66	-0.003	0.000	>50	2030-11-04	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2019-10-18	4.91	4.79	0.022	0.024	>50	2030-11-04	
3-CW-101-02	5.49	2.79	2019-10-18	4.98	4.69	0.025	0.057	33.1	2030-11-04	
3-CW-101-03	5.49	2.79	2019-10-18	4.87	4.63	0.031	0.048	38.7	2030-11-04	
3-CW-101-04	5.49	2.79	2019-10-18	5.20	5.22	0.008	-0.004	>50	2030-11-04	
3-CW-101-05	5.49	2.79	2019-10-18	5.14	5.19	0.010	-0.010	>50	2030-11-04	
3-CW-101-06	5.49	2.79	2019-10-18	5.01	4.91	0.019	0.020	>50	2030-11-04	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2019-10-18	4.85	4.77	0.025	0.016	>50	2030-11-04	
3-CW-102-02	5.49	2.79	2019-10-18	4.17	4.09	0.050	0.016	26.2	2030-11-04	
3-CW-102-03	5.49	2.79	2019-10-18	4.27	4.12	0.049	0.030	27.0	2030-11-04	
3-CW-102-04	5.49	2.79	2019-10-18	4.50	4.29	0.043	0.042	35.2	2030-11-04	
3-CW-102-05	5.49	2.79	2019-10-18	4.45	4.35	0.036	0.020	43.9	2030-11-04	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2024-11-07	_____